LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

ANSWER KEY & MARKING SCHEME

Ch 9 Test: Light – Reflection and Refraction

Chapter / Topic: Light – Reflection and Refraction

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SECTION A

Q1. Which of the following can make a parallel beam of light when light from a point source is incident on it? **[1]**

Answer: (d) Concave mirror and convex lens (with the source placed at the focus).

Q2. The image formed by a concave mirror is virtual, erect and larger than the object. Where is the object placed? **[1]**

Answer: (c) Between the pole P and F

Q3. The focal length of a lens is +50 cm. Its power is: **[1]**

Answer: (b) +2 D. P = 1/f (in metres) = 1/0.5 m = +2 D

Q4. The refractive index of glass is 1.5. The speed of light in glass is (speed of light in vacuum = 3×10^8 m/s): **[1]**

Answer: (b) 2×10^8 m/s. v = c/n = $3 \times 10^8/1.5$

Q5. The mirror used as a rear-view mirror in vehicles is: **[1]**

Answer: (d) convex. It gives an erect image and a wider field of view.

Q6. **Assertion (A):** The magnification produced by a convex mirror is always positive and less than 1. **Reason (R):** A convex mirror always forms a virtual,… **[1]**

Answer: (a)

SECTION B

Q7. Define the principal focus of a concave mirror. The radius of curvature of a spherical mirror is 20 cm. What is its focal length? **[2]**

Answer: Rays parallel to the principal axis, after reflection from a concave mirror, all meet at a point on the principal axis called the principal focus. f = R/2 = 20/2 = 10 cm.

- Definition: 1
- f = 10 cm: 1

OR, answer: (1) The incident ray, the refracted ray and the normal at the point of incidence all lie in the same plane. (2) For a given pair of media and a given colour of light, the ratio sin i / sin r is a constant (Snell's law).

- Each law: 1

Q8. A ray of light travelling in air enters water at an angle. Does it bend towards or away from the normal? Why? **[2]**

Answer: It bends towards the normal. Light travels more slowly in water, which is optically denser than air.

- Towards the normal: 1
- Reason (slower in denser medium): 1

SECTION C

Q9. An object 4 cm high is placed 15 cm in front of a concave mirror of focal length 10 cm. **[3]**

(i) u = −15 cm, f = −10 cm. 1/v = 1/f − 1/u = −1/10 + 1/15 = −1/30, so v = −30 cm. The image is 30 cm in front of the mirror. [2]

- Formula and correct signs: 1

• v = −30 cm: 1

(ii) m = −v/u = −(−30)/(−15) = −2, so h′ = −2 × 4 = −8 cm. The image is real, inverted and magnified (8 cm tall). [1]

Q10. A concave lens has a focal length of 15 cm. At what distance from the lens should an object be placed so that the image is formed 10 cm from the… **[3]**

Answer: f = −15 cm, v = −10 cm (a concave lens forms a virtual image on the same side). 1/v − 1/u = 1/f, so 1/u = 1/v − 1/f = −1/10 + 1/15 = −1/30, and u = −30 cm. The object is 30 cm from the lens. m = v/u = (−10)/(−30) = +1/3 ≈ 0.33. The image is virtual, erect and diminished.

- Formula and signs: 1
- u = −30 cm: 1
- m = +1/3 with nature: 1

OR, answer: u = −15 cm, f = +10 cm. 1/v = 1/f + 1/u = 1/10 − 1/15 = 1/30, so v = +30 cm (on the other side of the lens). m = v/u = 30/(−15) = −2. The image is real, inverted and magnified (twice the size).

- Formula and signs: 1
- v = +30 cm: 1
- m = −2 with nature: 1

SECTION D

Q11. Answer the following: **[5]**

(i) Diagram: one ray parallel to the principal axis passes through F_2 after refraction, and one ray through the optical centre goes straight without deviation. They meet beyond $2F_2$. Image: beyond $2F_2$; real, inverted and enlarged. [3]

- Correct ray diagram (two rays, arrows): 2
- Position and nature: 1

(ii) 1 dioptre is the power of a lens whose focal length is 1 metre. P = 1/f = 1/(−2 m) = −0.5 D. [2]

- Definition: 1
- P = −0.5 D: 1

OR, answer: (a) Concave: torches and vehicle headlights (parallel beams), shaving mirrors, dentists' mirrors, solar furnaces. Convex: rear-view mirrors in vehicles, mirrors at blind turns, security mirrors in shops. (b) u = −25 cm, f = +10 cm. 1/v = 1/10 − 1/25 = 3/50, so v = +16.7 cm. m = v/u = 16.7/(−25) ≈ −0.67, so h′ ≈ −3.3 cm. The image is real, inverted and diminished, 16.7 cm on the other side of the lens.

- (a) 2 uses each: 1 + 1
- (b) v: 1, size: 1, nature: 1

SECTION E

Q12. The refractive index of a medium tells us how much light slows down in it compared with a vacuum. Some values are: water 1.33, kerosene 1.44, crown… **[4]**

(i) Water, which has the lowest refractive index. [1]

(ii) Diamond [1]

(iii) $v = c/n = (3 \times 10^8)/2.42 \approx 1.24 \times 10^8$ m/s [2]

- Formula v = c/n: 1
- Answer with unit: 1